

Canadian Wildfires in 2023

Visualization of the data in an interactive Map

This notebook produces an interactive Folium visualization of the recorded uncontrolled wildfire events in Canada in 2023 and their impact on the most relevant cities in Canada.

Therefore it combines three togglable layers * **Heatmap** with all uncontrolled NFDB fires in 2023, weighted by size class

- **Cities as CircleMarkers** coloured by their fire-impact score
- **Live heatmap** of recent active fires retrieved from the NASA FIRMS API

Short summary of the following steps

0. Import & Constants
1. Computation of GeoDataFrames & CRS Reprojection
2. Spatial Join & Fire Scores for Cities
3. Classification of cities impact levels
4. Live fire data from NASA FIRMS
5. Put together the map layers
6. Export of the final map

IMPORTANT!! an active internet connection is necessary to run this notebook, since otherwise the NASA FIRMS data can't be loaded.

0. Import & Constants

```
import requests
from datetime import date
from pathlib import Path

import geopandas as gpd
import pandas as pd
import folium
from folium import plugins
import leafmap.foliummap as leafmap

DATA_DIR = Path("../data")

FIRE_CSV_PATH = DATA_DIR / "processed" / "fire_2023_clean.csv"
FIRE_GEOJSON_PATH = DATA_DIR / "processed" / "fire_2023_points.geojson"

CITIES_CSV_PATH = DATA_DIR / "processed" / "selected_cities_canada.csv"
CITIES_GEOJSON_PATH = DATA_DIR / "processed" / "selected_cities_canada.geojson"

LIVE_FIRE_PATH = DATA_DIR / "processed" / "fire_live.csv"

MAP_OUTPUT_PATH = Path("../outputs") / "interactive_firemap.html"
```

a) Analysis parameters

```
# EPSG:4326 in geographic degrees; required by Folium and for GeoJSON export.
# EPSG:3347 in metres; required for buffer & spatial join.

CRS_DEFAULT_DEGREE = "EPSG: 4326"
CRS_COUNTRY_METRIC = "EPSG: 3347"

#Storing the symbology characteristics to change it just once for both heatmaps
#HM means HeatMap.

HM_RADIUS = 13
HM_BLUR = 8
HM_MIN_OPACITY = 0.35

#inferno-inspired colour ramp, where dark > light means low > high intensity.
HM_GRADIENT = {
    0.00: "#000004",
    0.20: "#420A68",
    0.40: "#6A176E",
    0.60: "#932667",
    0.80: "#DD513A",
    1.00: "#FCFFA4",
}

#Later cities accumulate a fire score from number of fires within this radius
BUFFER_KM = 200

#Setup of fire_score thresholds which define for each city the impact level.
#S therefore means Score
S_LOW = 20
S_MODERATE = 50
S_HIGH = 256

#Essential variables for NASA FIRMS live fire API data from VIIRS NOAA-20 NRT data
FIRMS_MAPKEY = "b73850f35cfea7f219a3969b6dfd1d9a6"

CANADA_BBOX = {"north": 73.2267, "south": 44.6401, "west": -140.625, "east": -49.2188}

#For layering the three toggleable layers we need to set up first a basemap for Canada.
MAP_CENTER = [58.03,-90.82]
ZOOM_START = 4
TILE_STYLE = "CartoDB.DarkMatterNoLabels"
```

b) Load Processed Data

```
fire_2023      = pd.read_csv(FIRE_CSV_PATH)
cities_df      = pd.read_csv(CITIES_CSV_PATH)

assert len(fire_2023) > 0, "Failed; you may re-run the data cleaning notebook."
```

```
assert len(cities_df) > 0, "Failed; you may re-run the city selection notebook."

print(f"Loaded {len(fire_2023)} wildfire events and {len(cities_df)} cities of Canada")
```

Loaded 964 wildfire events and 40 cities of Canada

1. Computation of GeoDataFrames & CRS Reprojection

```
#For visualization in Folium we need the data in a GeoDataFrame and as a GeoJSON file
fire_2023_gdf = gpd.GeoDataFrame(
    fire_2023, geometry=gpd.points_from_xy(
        fire_2023.longitude, fire_2023.latitude),
    crs= CRS_DEFAULT_DEGREE)

fire_2023_gdf.to_file(FIRE_GEOJSON_PATH, driver="GeoJSON")

selected_cities_gdf = gpd.GeoDataFrame(
    cities_df,
    geometry=gpd.points_from_xy(
        cities_df.lng,
        cities_df.lat),
    crs= CRS_DEFAULT_DEGREE)

selected_cities_gdf.to_file(CITIES_GEOJSON_PATH, driver="GeoJSON")

#For operations based on distance metric projection is required
cities_metric_gdf = selected_cities_gdf.to_crs(CRS_COUNTRY_METRIC)
fire_2023_metric_gdf = fire_2023_gdf.to_crs(CRS_COUNTRY_METRIC)
```

2. Spatial Join & Fire Scores for Cities

a) Setting up buffers around cities

```
#Buffer the city to then count the fires inside it for impact score
cities_buffered = cities_metric_gdf.copy()
cities_buffered["geometry"] = cities_buffered.geometry.buffer(
    BUFFER_KM * 1000)
```

b) Spatial Join

```
#A Spatial Join is needed to find all fires inside the buffer
#Since we want to have also cities with no fires within their buffer
#we make a left join.

joined_data = gpd.sjoin(
    cities_buffered,
    fire_2023_metric_gdf,
```

```

    how="left", #cities without fires should be visible
    predicate = "contains"
)

```

c) Aggregation fire score per city

```

#Aggregating fire scores per city to gain an impact score.
#The class_weight was set up to consider the severity.

fire_per_city = (
    joined_data
    .groupby("city")["class_weight"]
    .sum()
    .reset_index()
    .rename(columns={"class_weight": "fire_score"})
)

#Merging the scores back gives us all relevant data together and by
#converting it into the geographic CRS it can be later visualized.

selected_cities_gdf = (
    cities_metric_gdf
    .merge(fire_per_city, on="city", how="left")
    .to_crs(CRS_DEFAULT_DEGREE)
)

#For replacing the NaNs (Cities with no fires in their buffer)
#we fill it up with a score of 0

selected_cities_gdf["fire_score"] = selected_cities_gdf[
    "fire_score"].fillna(0)

```

3. Classification of cities impact levels

```

def impact(fire_score):
    """
    Classifies the cumulative fire score of a city into wildfire impact

    The thresholds are defined by the constants SCORE_LOW,
SCORE_MODERATE, and SCORE_HIGH. Those are the sums of fire class_weight
values (1 / 2 / 4 / 16 / 256) from the NFDB cleaning step.

    Parameters
    -----
    fire_score_1
        Cumulative class_weight of all fires within CITY_BUFFER_M of a city.

    Returns
    -----
        One of: "No impact", "Low impact", "Moderate impact", "High impact", "Very high impact".
    """

```

```

"""
if fire_score == 0:
    return "No impact"
if fire_score < S_LOW:
    return "Low impact"
if fire_score < S_MODERATE:
    return "Moderate impact"
if fire_score < S_HIGH:
    return "High impact"
return "Very high impact"

selected_cities_gdf["impact"] = selected_cities_gdf[
    "fire_score"].apply(impact)

print(selected_cities_gdf["impact"].value_counts())

```

```

impact
No impact      13
Low impact     11
Very high impact  6
Moderate impact  6
High impact     4
Name: count, dtype: int64

```

4. Live fire data from NASA FIRMS

```

today = date.today()
api_url = f"https://firms.modaps.eosdis.nasa.gov/usfs/api/area/csv/{FIRMS_MAPKEY}/VIIRS_NOAA20_NRT/world"

response = requests.get(api_url)

if response.status_code == 200:
    fire_live_df = pd.read_csv(api_url)
    fire_live_df.to_csv(LIVE_FIRE_PATH, index=False)

    fire_live_df = fire_live_df[
        (fire_live_df["latitude"] >= CANADA_BBOX["south"]) &
        (fire_live_df["latitude"] <= CANADA_BBOX["north"]) &
        (fire_live_df["longitude"] >= CANADA_BBOX["west"]) &
        (fire_live_df["longitude"] <= CANADA_BBOX["east"])
    ]
    print(f"Live fire records for Canada: {len(fire_live_df)}")
else:
    print("Error")

```

Live fire records for Canada: 32

5. Put together the map layers

a) Base map

```
#By using leafmap we make sure that we can add a legend.
canada_map = leafmap.Map(
    location = MAP_CENTER,
    zoom_start = ZOOM_START,
    tiles = TILE_STYLE,
)
```

b) Weighted Heat Map for wildfire events in 2023

```
#Heatmap for weighted fires
heat_data = [[point.xy[1][0], point.xy[0][0], weight]
              for point, weight in zip (
                  fire_2023_gdf.geometry,
                  fire_2023_gdf["class_weight"])
              ]

plugins.HeatMap(
    heat_data,
    radius=HM_RADIUS,
    blur=HM_BLUR,
    min_opacity=HM_MIN_OPACITY,
    name = "Wildfires in 2023",
    gradient= HM_GRADIENT
).add_to(canada_map)
```

```
<folium.plugins.heat_map.HeatMap at 0x232affd3b60>
```

c) Visualization of impact on selected cities

```
impacted_cities_layer = folium.FeatureGroup(name = "Impacted Cities in 2023")

for city,row in selected_cities_gdf.iterrows():
    fire_score = row["fire_score"]
    if fire_score == 0:
        city_color = "#E0FFFF"
        city_radius = 4

    elif fire_score > 0 and fire_score < 20:
        city_color = "#87CEFA"
        city_radius = 5

    elif fire_score >= 20 and fire_score < 50:
        city_color = "#1E90FF"
        city_radius = 7
```

```

elif fire_score >= 50 and fire_score < 256:
    city_color = "#0000FF"
    city_radius = 8

else:
    city_color = "#191970"
    city_radius = 10

folium.CircleMarker(
    location=[row.geometry.y, row.geometry.x],
    radius = city_radius,
    color = "gray",
    weight = 0.9,
    fill = True,
    fill_color=city_color,
    fill_opacity = 0.95,
    tooltip= (f"Name: {row['city']} <br>"
              f"Country: {row['country']}<br>"
              f"Population: {row['population']} <br>"
              f"Impact: {row['impact']}"),
    ).add_to(impacted_cities_layer),

impacted_cities_layer.add_to(canada_map)

```

<folium.map.FeatureGroup at 0x232b00fd6a0>

d) HeatMap for NASA FIRMS recent data

```

live_heatmap = [
    [row["latitude"], row["longitude"]]
    for _, row in fire_live_df.iterrows()
]

plugins.HeatMap(
    live_heatmap,
    radius=HM_RADIUS,
    blur=HM_BLUR,
    min_opacity=HM_MIN_OPACITY,
    name = "Live wildfire",
    show = False,
    gradient= HM_GRADIENT
).add_to(canada_map)

```

<folium.plugins.heat_map.HeatMap at 0x232b002b250>

e) Legend and Layer control

```

legend_impact ={
    "No Impact": "#E0FFFF",

```

```

    "Low Impact": "#87CEFA",
    "Medium Impact": "#1E90FF",
    "High Impact": "#0000FF",
    "Very High Impact": "#191970"
}

canada_map.add_legend(title="Fire impact on city",
                      legend_dict=legend_impact)

folium.LayerControl(collapsed=False).add_to(canada_map)

```

```
<folium.map.LayerControl at 0x232b1714980>
```

6. Export of the final map

```

canada_map.save(MAP_OUTPUT_PATH)

print(f"Map saved to: {MAP_OUTPUT_PATH}")

```

```
Map saved to: ..\outputs\interactive_firemap.html
```

Display Map

```

#Visualization of Impact Map for Wildfire season 2023 with togglable layers
canada_map

```